0.1 Overview

The NA62 experiment is a fixed target experiment located in the north area of CERN with a high energy particle beam provided by the Super Proton Synchrotron (SPS). The main goal of the experiment is to measure the branching fraction of the ultra-rare charged kaon decay $K^+ \to \pi^+\nu\bar{\nu}$. Alongside, there is an expansive physics program of precision and rare decay measurements, as well as searches for forbidden processes and exotic particles.

The experiment employs a decay-in-flight technique to obtain a large number of charged kaon decays, in the order of 10^{13} within the short time frame of a few years. The experimental setup was designed to fulfil all the requirements for the challenging measurement of $K^+ \to \pi^+\nu\bar{\nu}$, which involves the detection of the K^+ upstream and the detection of the product π^+, with a missing energy-momentum carried away by the undetected neutrino-antineutrino pair. The measurement of the missing mass is used for kinematic rejection of the main background K^+ decays; $K^+ \to \pi^+\pi^0$, $K^+ \to \pi^+\pi^-$ and $K^+ \to \mu^+\nu_\mu$ by selecting two regions shown in figure 1. The missing mass calculation requires precise timing and position measurement of the K^+ and π^+ candidates, provided by the beam (section 0.3.2) and secondary (section 0.3.4) particle spectrometers. The measured 3-momenta of the candidates allows for the reconstruction of the 4-momenta of the Kaon P_{K^+} and the pion P_{π^+} which is used in equation 0.1

$$m^2_{miss} = (P_{K^+} - P_{\pi^+})^2 \tag{0.1}$$

The rest of this chapter will cover how NA62 fulfils the above-mentioned requirements. Section 0.2 will describe the NA62 beam-line from the target through the NA62 detector. The NA62 detector comprises a number of sub-detectors, which will be described in Section 0.3, with the trigger and data acquisition system (TDAQ) following in Section 0.4

0.2 The NA62 Beam Line

The NA62 experiment at CERN uses a hadron beam extracted from the SPS with a nominal momentum of 400 GeV/c, directed onto a 400 mm long, 2 mm diameter

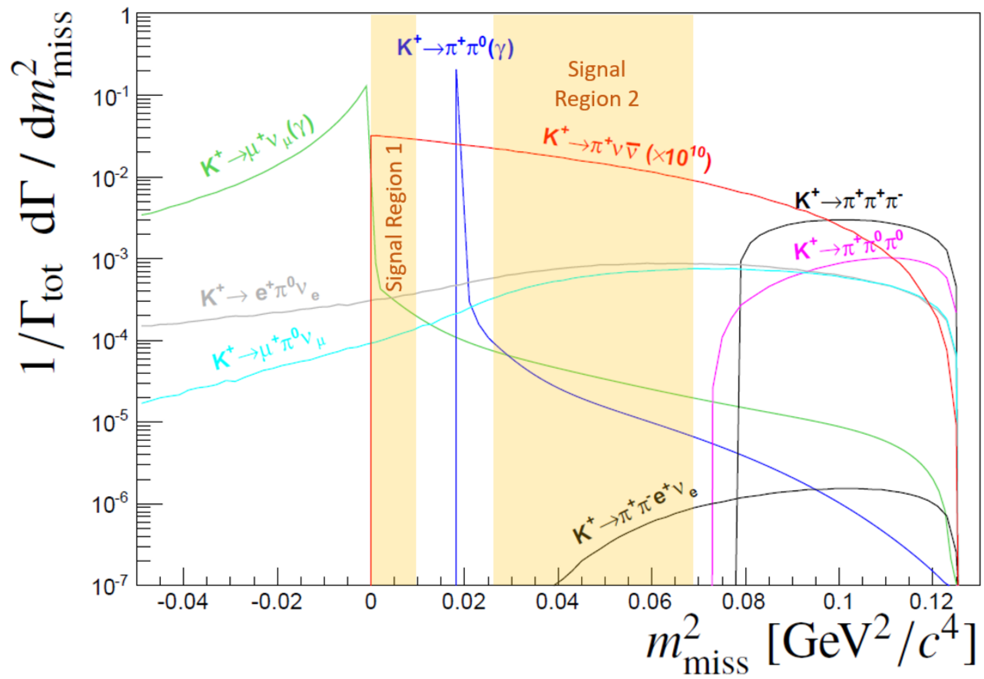

Figure 1: Expected squared missing mass distributions of main kaon decays and $K^+ \to \pi^+\nu\bar{\nu}$ (scaled by 10^{10} for for visibility). The solid coloured areas show the signal regions used to reject the dominant kaon decays kinematically [1].

beryllium rod that forms the fixed target (T10) located at 0 m in figure 3. The beam derived from the T10 target, referred to as K12, is a high-intensity unseparated hadron beam with a nominal momentum of 75 GeV/c. The beam momenta was chosen to maximise the fraction of kaons with respect to the other beam components at the entrance to the detector.

The elements of the K12 beam-line are shown in Figure 3 and comprise a number of components. First are three quadrupole magnets (Q1, Q2 and Q3), used to focus the beam with a large solid angle acceptance of ±2.7 mrad horizontally and ±1.5 mrad vertically at a 75 GeV/c central momentum. Next follows an achromat (A1), which selects a beam of a 75 GeV/c momentum with a 1 % rms momentum. The A1 achromat comprises four dipole magnets that vertically deflect the beam, the first two deflect the beam 110 mm downwards, while the final two deflect the beam upwards towards the original axis. Between these pairs of magnets, there are a pair of water-cooled beam-dump units (TAX1 and TAX2) with a set of holes to select the required 75 GeV/c momentum beam while absorbing unwanted secondary beam particles and any remaining primary protons. TAX1&2 can be closed to act

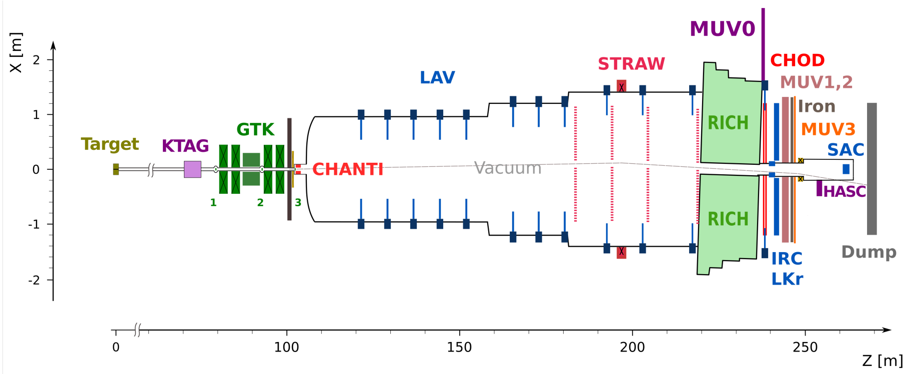

Figure 2: X-Z diagram of NA62 detector setup for Run 1.

as a secondary target for use in dump mode or for safety when access is required downstream. At the focus between the TAX1&2, a radiator consisting of tungsten plates with a thickness of up to 5 mm is introduced into the path of the beam. The radiator is optimised to reduce positron energy while minimising the loss of energy to hadrons. A triplet of quadrupoles (Q4, Q5, Q6) follows, which refocuses the beam in the vertical dimension and produces a parallel beam in the horizontal dimension. The space between the quadrupoles is occupied by a pair of collimators (C1, C2), which redefines the horizontal and vertical acceptance of the beam. The collimator after the Q6 quadrupole (C3) is employed to absorb positrons that had their momenta degraded by the radiator and again redefines the acceptance in the vertical dimension.

The beam next passes through a 40 mm field-free hole in a set of iron plates, placed between three 2 m dipole magnets (B3). The vertical magnetic field in the iron plates is used to deflect muons. Any deviation in the beam due to stray fields in the beam hole is cancelled out by a pair of steering dipoles (TRIM2 and TRIM3). Two quadrupoles (Q7, Q8) follow and ensure that the beam is parallel before the KTAG, which employs a differential Cherenkov counter (CEDAR), described in Section 0.3.1, which requires the beam to be parallel. Before the KTAG, a pair of collimators (C4 and C5) are used to absorb particles in the tails of the beam.

For beam monitoring, two pairs of filament scintillator counters (FISC 1, 3 and FISC 2, 4) are installed at either end of the KTAG. Used in coincidence, the beam divergence can be measured and used to tune the divergence to zero and ensure the beam is parallel through the CEDAR vessel.

Following the KTAG are two quadrupoles (Q9 and Q10), used to focus the beam

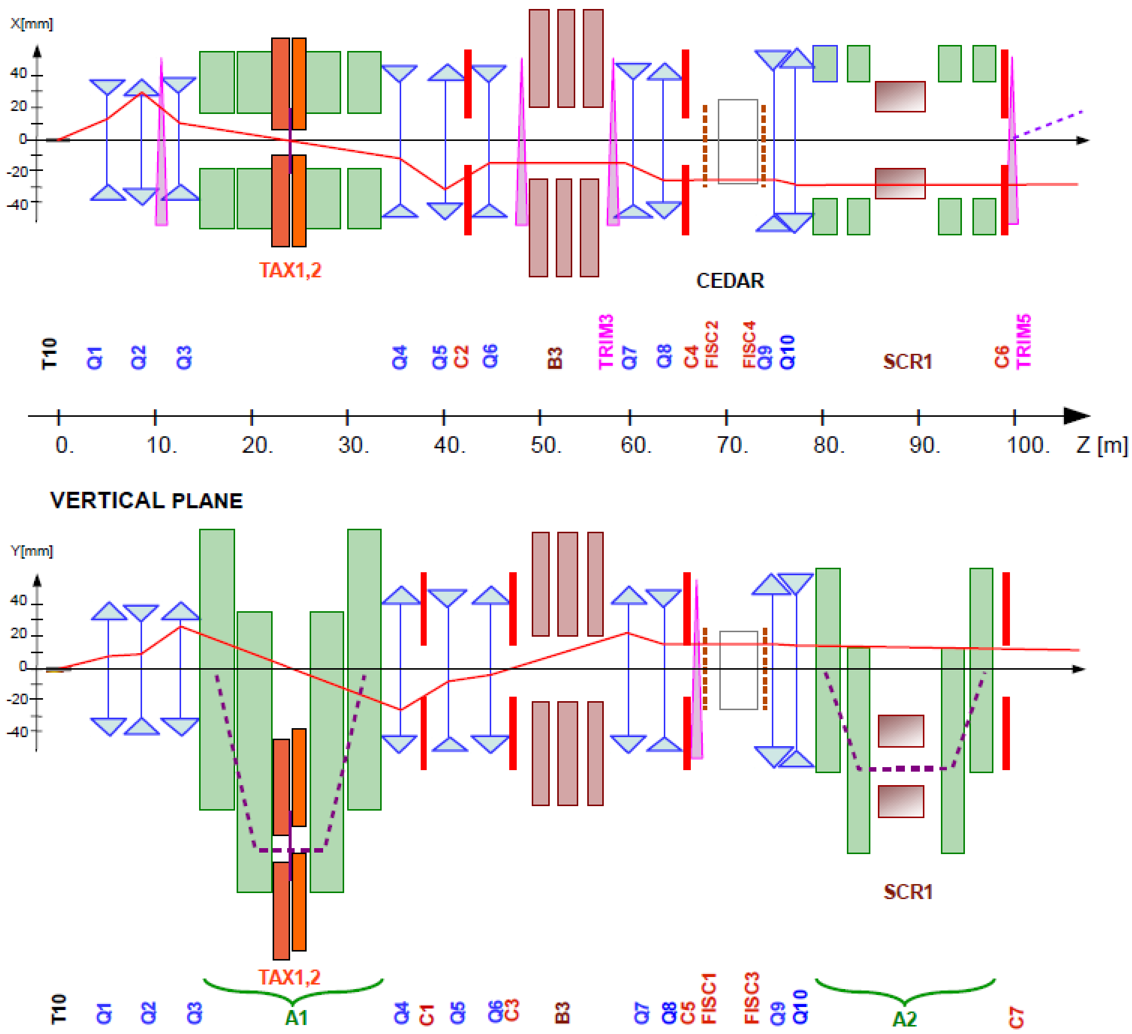

Figure 3: Schematic of K12 beamline from T10 target at 0m to the entrance of the NA62 decay region at 102.4m [1].

before the beam tracking system (GTK). The GTK comprises four stations made up of silicon pixel detectors described in Section 0.3.2 and are installed in the beam vacuum. The GTK stations are installed so that the space between GTK1 and GTK3 is occupied by a second achromat (A2), which comprises four dipole magnets that deflect the beam 60 mm vertically, allowing for the beam particle momenta to be measured. The return yokes of the third and fourth magnets and a magnetised iron collimator (SCR1) defocus beam muons, which are ejected from the beam between the second and third magnets. A pair of cleaning collimators (C6 and C7) used to absorb beam particles outside of the acceptance are located before the final GTK station (GTK3), which is at the entrance to the decay region ($Z = 102.4\,\mathrm{m}$). A steering magnet (TRIM5) is then used to deflect the beam in the horizontal dimension by an angle of +1.2 mrad.

At Z = 102.4 m begins a large 117 m vacuum pipe (Bluetube), evacuated to a pressure of 10^{-6} mbar by several cryo pumps along its length. The Bluetube is made up of 19 cylindrical sections ranging in diameter shown in Figure 2, from 1.92 m in the first section after GTK3 to 2.4 m in the middle section and then 2.8 m in the region containing the STRAW spectrometer. The first section is defined as the decay region or Fiducial Volume (FV) between Z = 105 and Z = 170 m. Further downstream, the Bluetube hosts a number of detectors: Eleven LAV stations used in the Photon detection system described in Section 0.3.6 and four STRAW spectrometer chambers coupled with a large aperture dipole magnet (MNP33), providing kinematic measurements of the charged products of kaons decaying in the decay region, described in Section 0.3.4. The MNP33 magnet provides a 270 MeV/c momentum kick in the horizontal dimension, deflecting the 75 GeV/c beam by -3.6 mrad, so that combined with the TRIM5 deflection, the beam passes through the central aperture of the Liquid Krypton calorimeter (LKr). The path of the beam is shown in Figure 4, showing the deflection in the horizontal dimension. After the final STRAW spectrometer chamber (STRAW4), the Bluetube is closed off with a thin aluminium window (except for the hole for the beam to pass through), separating the vacuum from the 17 m long, neon-filled Ring Imaging Cherenkov counter (RICH), designed to distinguish between secondary pions and muons produced in kaon decays and is part of the Particle Identification system (PID) and is described in Section 0.3.5. The beam is transported through the remaining downstream detectors in a vacuum and is deflected -13.2 mrad in the negative direction by a dipole magnet (BEND) to sweep the beam away from the Small Angle Calorimeter (SAC) acceptance, located within the vacuum beam pipe. Finally, the beam is absorbed in a beam dump.

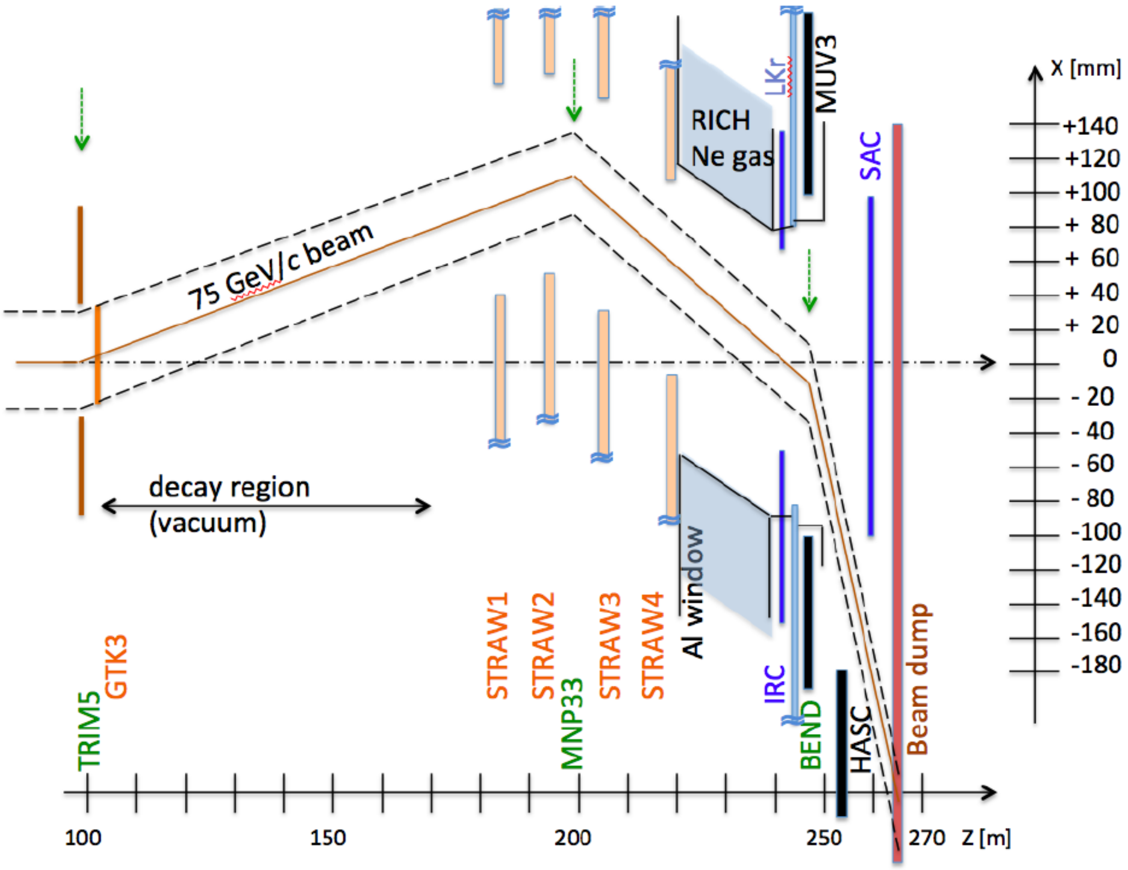

Figure 4: Schematic of K12 beamline after GTK3 [1].

0.3 NA62 Detectors

0.3.1 KTAG

As the NA62 beam is unseparated and kaons only account for 6 % of the beam composition, the kaons must be identified. Kaon identification is provided by the KTAG and located at $Z = 69.4$ m. It comprises a CEDAR vessel and a purpose-built photon detection system. Until the NA62 2023 run, the CEDAR vessel used was a CEDAR-W filled with a Nitrogen radiator. CEDARs were developed at CERN in the 1970s [2] for secondary beam particle detection at the super proton synchrotron (SPS). The CEDAR is a differential Cherenkov Counter and exploits the dependence of the beam particle's mass on the properties of the produced light, defined in equation 0.2 where $\beta = \frac{p}{\gamma m}$.

$$cos(\theta) = \frac{1}{\beta n} \tag{0.2}$$

In a beam of set momentum and a radiator with a constant refractive index, the angle of Cherenkov light is only affected by the mass of the beam particle. The pressure is set at 1.7 bar of Nitrogen in the CEDAR-W so that the light from the kaon is focused by a set of lenses and mirrors onto arrays of photomultiplier tubes (PMTs). The light from the other beam components does not reach the PMT arrays.

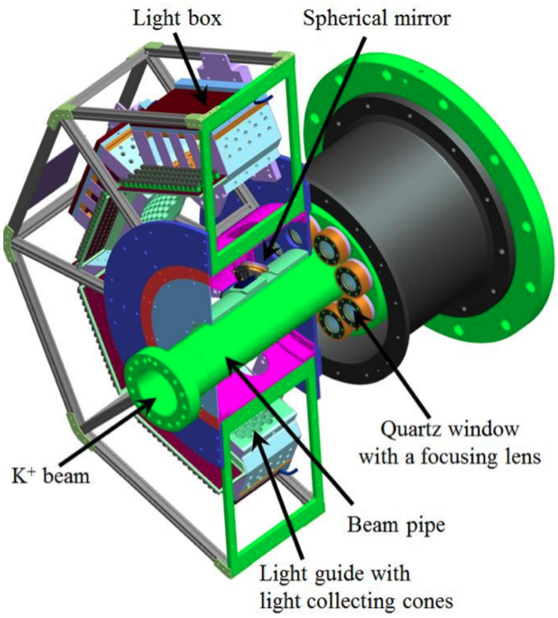

Figure 5: Model of the upstream part of the CEDAR and the KTAG photon-detection system [1].

In the original CERN design, the Cherenkov light exiting each CEDAR exit window was detected by a single ET-9820QB photomultiplier tube (PMT) placed directly on the exit window. A particle was identified if there was a coincidence of at least six PMTs. However, these PMTs would not fulfil the timing (≤ 100 ps) and rate requirements (45 MHz of kaons) at NA62.

For NA62, a purpose-built photon-detection system shown in Figure 5 was designed, constructed and mounted to the upstream end of the CEDAR Vessel. The photon light from the CEDAR exit windows is reflected radially 90 degrees by eight spherical mirrors onto eight PMT planes, referred to as sectors. Each PMT plane is equipped with an array of 48 single-anode Hamamatsu PMTs of two types: 32 R9880-110 and 16 R7400. The Quantum Efficiency (QE) for both Hamamatsu PMTs are shown in Figure 6. The QE is the probability of a PMTs photocathode to convert a photon of a specific wavelength and emit a photo-electron into the vacuum of the PMT.

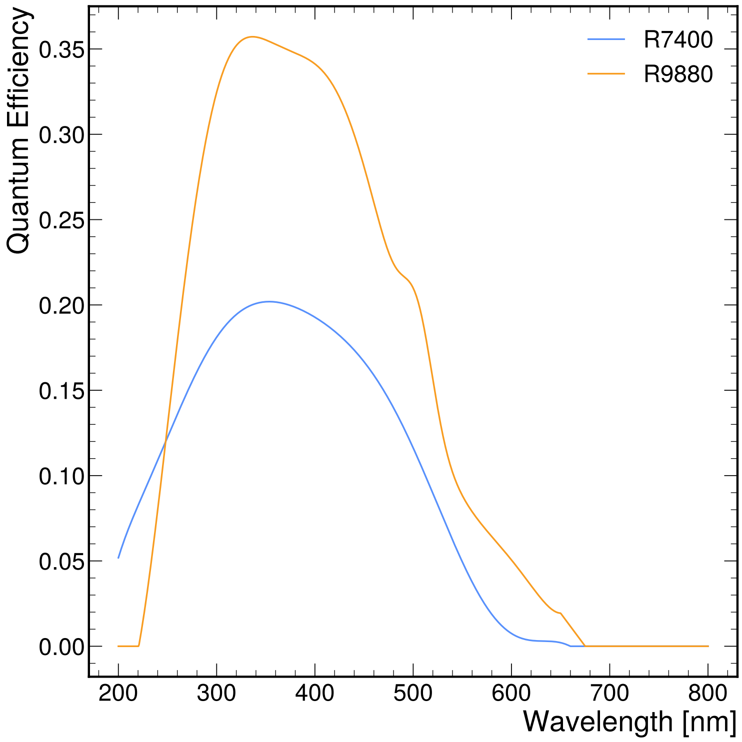

Figure 6: Quantum Efficiency of both types of PMT used in the KTAG photon-detection system (Hamamatsu R9880-110 and R7400).

The signals from the PMT arrays are clustered into kaon candidates within a nominal 2 ns time window. A kaon was identified when there are at least five sectors in coincidence (5-fold coincidence). The clustering results with, on average, 18 photo-electrons per kaon candidate. The PMTs can achieve a single hit time resolution of 300 ps, resulting in a reconstructed kaon time resolution of 70 ps and a kaon identification efficiency of 95% was achieved with the KTAG and CEDAR-W [3].

In 2023, a new CEDAR with a Hydrogen radiator (CEDAR-H) was installed in the NA62 beam-line after design at the University of Birmingham and testing at CERN. The new CEDAR uses the same basic principles as previous CEDARs but uses hydrogen as a radiator gas. This required the design of new optical elements in the CEDAR and the photon detection system. A more detailed description of the CEDAR and its optical elements, along with the design, testing and installation of the new CEDAR-H, can be found in Section **??**.

0.3.2 GigaTracker (GTK)

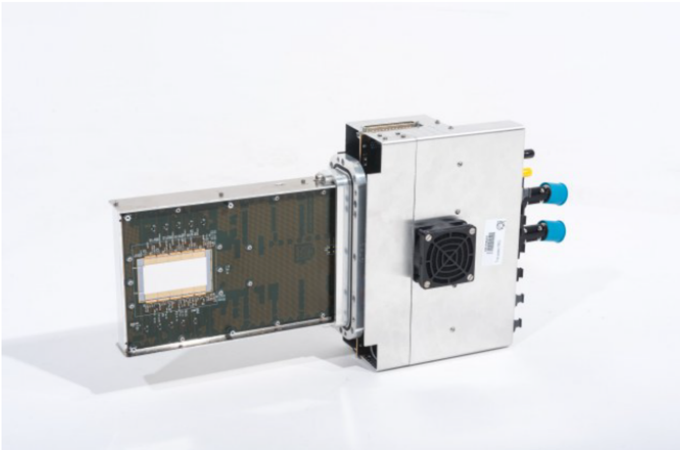

Figure 7: Image of GTk station [1].

The precise measurement of the beam momentum, time, and position is provided by the GigaTracker (GTK). A beam spectrometer comprising hybrid silicon trackers is placed within the vacuum pipe and an achromat upstream of the decay volume. As particles of differing momenta are deflected by different amounts in a magnetic field, the momentum of a particle can be measured by comparing the positions of the particles in each station. The detector was designed to measure a particles momentum at a resolution of 0.2 % and its direction at the exit of the achromat to 16 μrad. Each GTK station has a time resolution of 200 ps, which results in an overall track time resolution of 75[4]. The layout of the GTK is shown in Figure 8

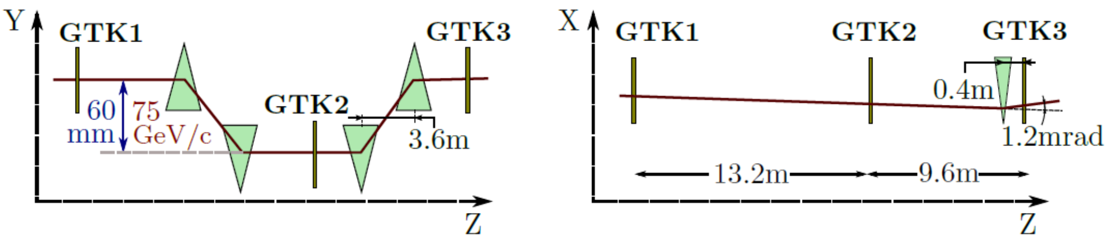

Figure 8: Schematic of GTK stations within the achromat for run 1 [1].

Each station consists of 18000 pixels of size $300 \times 300\,\mu m^2$ in a matrix of 200×90, covering a total area of $62.8 \times 27\,mm^2$ and is read out by application-specific integrated circuits (ASIC). The tracker is placed directly within the beam, which requires the material to be minimised in order to reduce the number of interactions with the beam and reduce the beam divergence. The material budget for each station was set at 0.5 %, corresponding to 500 μm of Silicon.

0.3.3 Charged Anti-coincidence detector (CHANTI)

The CHANTI is located after the GTK, and its primary goal is to provide the rejection of particles produced from inelastic collisions of beam particles with the final GTK station. Vetoing these particles is imperative as they would enter the decay volume and result in background for $K^+ \rightarrow \pi^+ \nu \bar{\nu}$. The detector is also used to detect the muon halo close to the beam. The CHANTI detector is a set of six square hodoscope stations comprising two planes, one paced vertically and the other horizontally, with a fully constructed station shown in Figure 9. The planes are comprised of polystyrene based scintillator bars with central fast wavelength-shifting (WLS) fibre which were then connected to silicon photomultipliers (SiPMs).

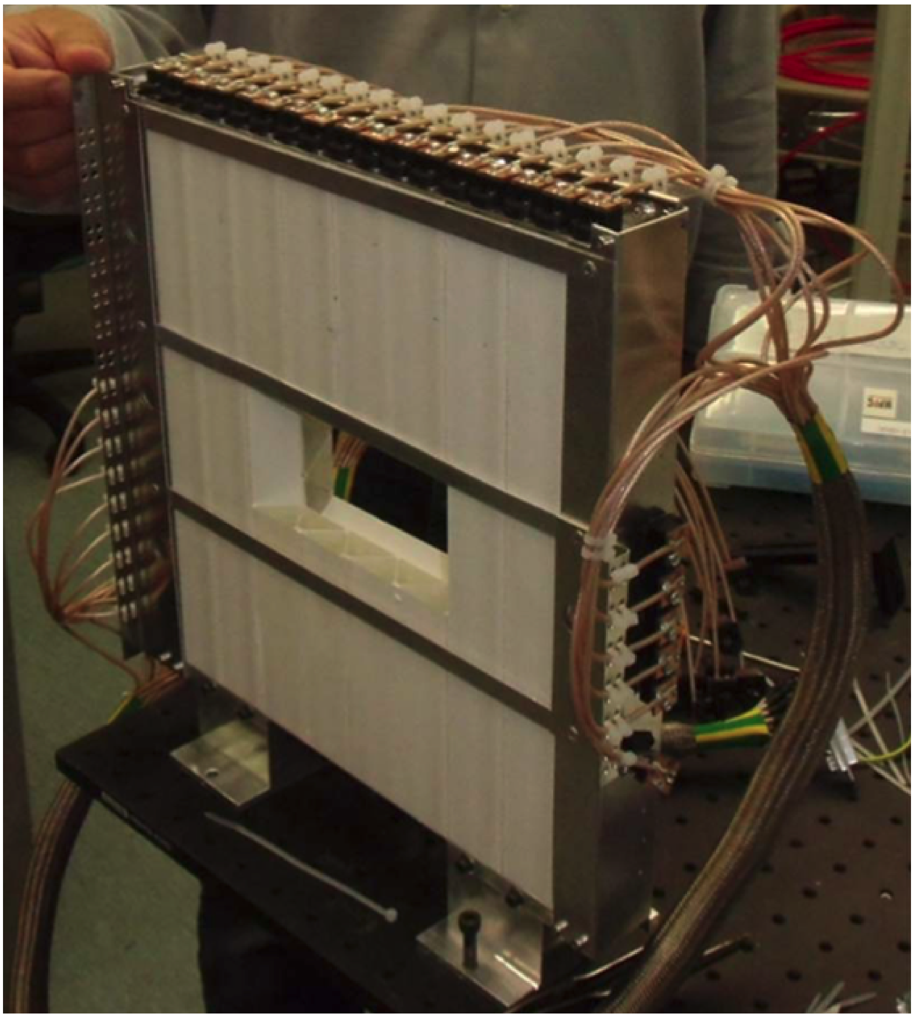

Figure 9: Image of a single CHANTI station [1].

0.3.4 Straw Spectrometer (STRAW)

The STRAW Spectrometer provides kinematic measurements of charged decay products produced in K^+ decays within the decay volume. It comprises four chambers and a large aperture dipole magnet (MNP33), providing measurement of secondary particle momentum with a resolution of

$$\frac{\sigma(p)}{p} = 0.30\% \oplus 0.005\% \tag{0.3}$$

where p is the track momentum in GeV/c. The chambers are downstream from the decay region, with the first chamber located at $Z = 183\,\mathrm{m}$ and the second 10 m further downstream. The MNP33 magnet is situated downstream of chamber two at $Z = 197.6\,\mathrm{m}$ and provides an integrated field of 0.9 Tm. Chambers 3 and 4 are located downstream of the MNP33, $Z = 204\,\mathrm{m}$ and 219 m respectively.

Each chamber comprises four views (X, Y, V, and U) rotated at 0 deg, 90 deg, +45 deg, and −45 deg, each consisting of four layers of straws shown in Figures 10 and 11. The views have a gap near the center of around 12 cm with no straws, such that when they are combined to form a chamber, an octagonal hole with an apothem of 6 cm is produced for the beam to pass through. As shown in Figure 4, the beam hole for each chamber is not centred and instead follows the deflection of the beam before and after the MNP33 magnet, with the beam hole parameters defined in Table 1.

Chamber	**Beam Hole Size (cm)**	**Beam Hole Offset (X, Y, Z)**
1	6	101.2 mm, 0 mm, 183 m
2	6	114.4 mm, 0 mm, 193 m
MNP33	-	-, -,198 m
3	6	92.4 mm, 0 mm, 204 m
4	6	52.8 mm, 0 mm, 218 m

Table 1: Straw spectrometer chamber beam-hole positions

The straws are 2.1 m long drift tubes that are 9.82 mm in diameter, made from 36 µm thick polyethylene terephthalate (PET), coated on the inside with 50 nm of copper and 20 nm of gold. The straws are filled with a mixture of 70 % Argon and

30 % CO_2 at atmospheric pressure and have a central gold-plated tungsten anode wire. The straws act like proportional counters, when a charged particle passes through the gas, it is ionised, creating electron-ion pairs that drift to the opposite electrodes, inducing a charge on the anode wire. When the electron approaches the anode wire, the electric field strength increases dramatically, producing Townsend avalanches. This provides a large multiplication of the signal, producing a charge pulse, which can be read out by the detector electronics.

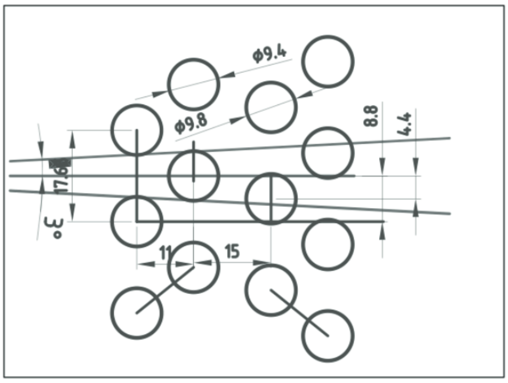

Figure 10: Diagram of straw placement in each view [5]

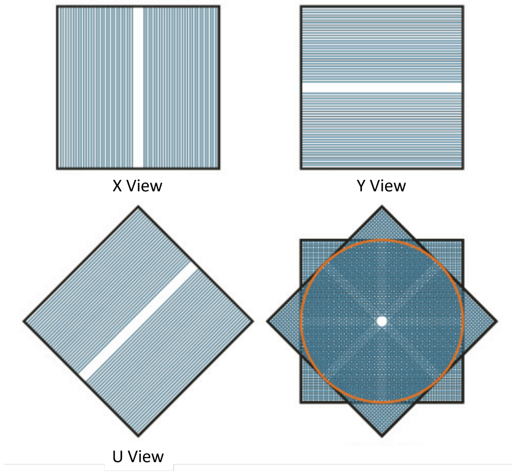

Figure 11: Diapgram showing placement of views in a straw chamber. Edited from [5]

0.3.5 Ring Imaging Cherenkov counter (RICH)

The Ring Imaging Cherenkov counter (RICH) is designed to separate secondary pion and muons within a momentum range of 15 - 35 GeV/c. The ability to separate the secondary particles is required as the largest background to $K^+ \to \pi^+\nu\bar{\nu}$ comes from the $K^+ \to \mu^+\nu_\mu$, which has a branching fraction 10 orders of magnitude greater. The majority of the suppression comes from kinematic selections and the response difference between pions and muons in the calorimeters. However, another factor of 100 is required in the momentum range 15 - 35 GeV/c, which the RICH satisfies.

The RICH is a 17.5 m long cylindrical vessel comprising four steel sections with diameters gradually decreasing from 4.2 m at the upstream section to accommodate the PMT flanges to 3.2 m in the downstream end. The radiator vessel is separated from the vacuum decay tube by an aluminium window of 2 mm thickness and radius of 1.1 m at the upstream end, allowing minimal material in the path of particles entering the detector. A lightweight aluminium pipe connected to the vacuum tube passes through the vessel, allowing the beam to pass aluminium in a vacuum and continues through the 4 mm thick, 1.4 m radius aluminum exit window. The vessel contains Neon gas at a pressure of 900 mbar with a refractive index of $n = 1 + 62.8 \times 10^{-6}$ at a wavelength of 300 μm, corresponding to a Cherenkov threshold for charged pion of 12.5 GeV/c.

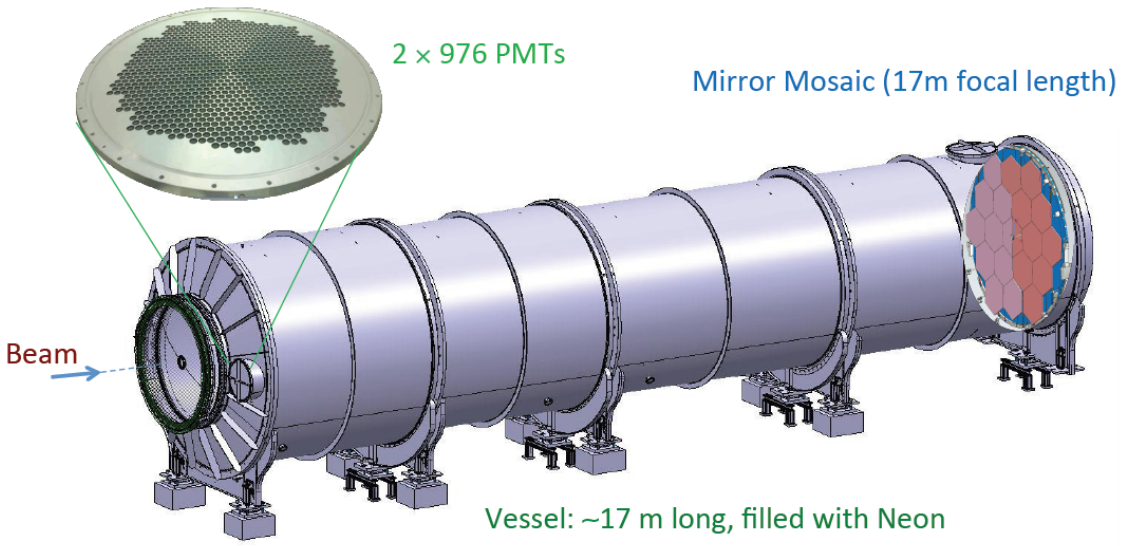

Figure 12: Schematic of RICH detector, showing the beam entering the upstream end on the left. A zoomed-in image of one of the two PMT flanges shows the layout of the PMT arrays. The mirrors are also shown at the downstream end of the vessel [6]

A mosaic of 20 spherical mirrors (18 hexagonal and two semi-hexagonal) with a focal length of 17 m are located at the downstream end of the vessel and is shown in Figure 12. This mirror system reflects and focuses the Cherenkov light toward two PMT arrays located at the upstream end of the vessel. The two PMT arrays are located on either side of the entrance window (see Figure 12), each consisting of 976 Hamamatsu R7400-U03 PMTs. Each PMT has an 8 mm active area and are packed into a hexagonal lattice, each with a Winston cone directing light onto the active area.

The particles are identified (selected or rejected) using a cut on the particle's mass, calculated from the particle velocity using the reconstructed Cherenkov ring radius shown in Figure 13 and the momentum of the particle measured by the spectrometer. In Figure 13, it is clear that separating pions and muons within the desired momentum range is possible.

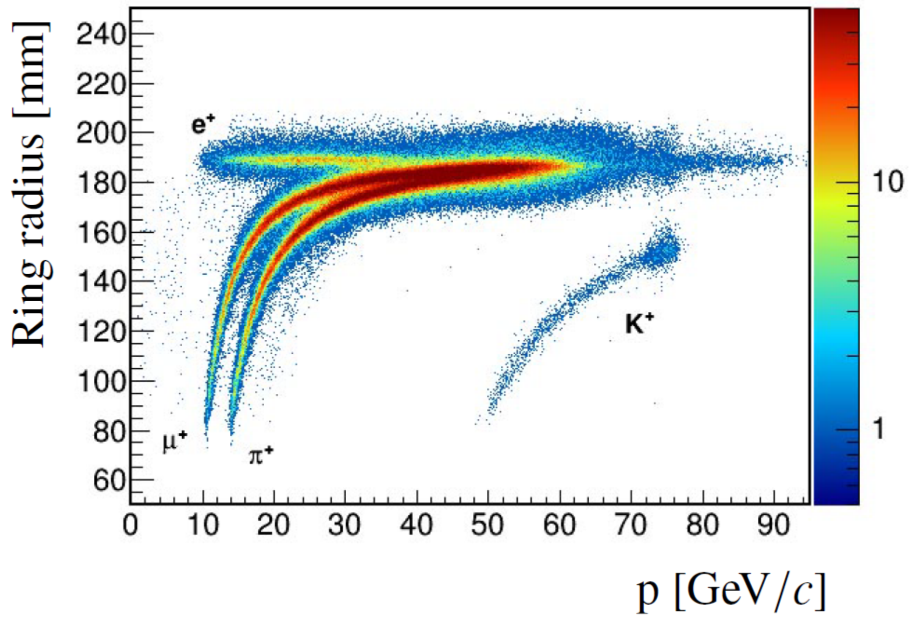

Figure 13: Cherenkov ring radius as a function of the particle momentum [6]

0.3.6 Photon Detection System

The photon detection system provides a rejection power of 10^8 for the main decay background $K^+ \rightarrow \pi^+\pi^0$. It is made up of four calorimeters used to detect and reject background decays with photons in the final state, with coverage from 0 to 50 mrad. Photons with small production angles between 0 and 1 mrad is covered by the intermediate-ring calorimeter (IRC) and small-angle calorimeter (SAC). The IRC wraps around the beam pipe and consists of layers of lead and scintillating material, which are read out by WLS fibres bundled into four groups and read out by a Hamamatsu PM. The SAC, which is located just upstream of the beam dump within the beam vacuum, comprises of 70 lead plates with 70 plastic scintillators in between readout by WLS fibres resulting in a total material of 19 X_0. The fibres are bundled into four groups which are then read out by Hamamatsu PMs, resulting in four channels.

Photons with production angles between 1 and 8.5 mrad are covered by the NA48 liquid krypton calorimeter (LKr). A quasi-homogeneous calorimeter which consists of a large cryostat containing 9000 litres of liquid krypton, which extends from the beam-pipe to a radius of 128 cm and has a depth of 127 cm. This corresponds to a radiation length of 27 X_0, designed to allow for the full capture of electromagnetic showers produced. Thin beryllium-copper ribbons make up the 13248 longitudinal cells, depicted in Figure 14, consisting each of a central anode and cathodes on either side. Each anode ribbon was connected at the downstream end to preamplifiers and the high voltage of 3000V to provide the drift field required for the ionisation.

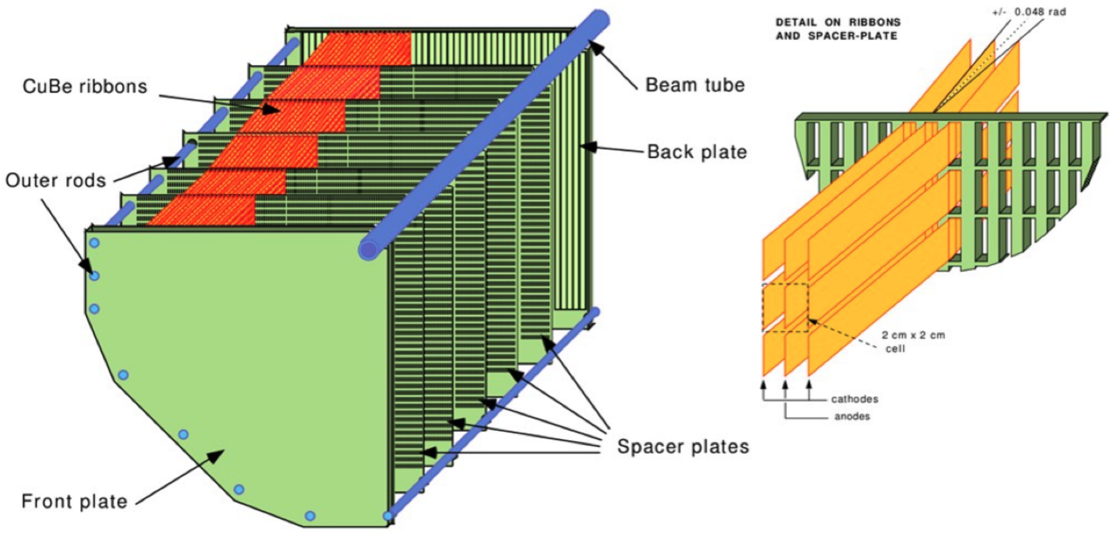

Figure 14: Left, a schematic of the structure of a single quadrant of the Lkr. Right, the layout of cells within the quadrant. [6]

The Large Angle Veto (LAV) provides coverage from 8.5 to 50 mrad, with 12 stations positioned along and beyond the decay volume. Eleven of the LAV stations are along the vacuum volume with the final station placed just upstream of the LKr. A LAV station consists of a barrel of lead glass blocks surrounding the perimeter of the vacuum volume and was recycled from the OPAL experiment [7], with LAV1 picture in Figure 15. The electromagnetic showers produced in each of the glass blocks result in the emission of Cherenkov light, which is then detected by a Hamamatsu R2238 76 mm diameter photomultiplier attached to the back side.

Figure 15: Photo of the installation of LAV1 at NA62 in 2008.

0.3.7 Charged Particle Hodoscopes

The charged particle hodoscopes comprise a pair of scintillating detectors which cover the acceptance downstream of the RICH and upstream of the LKr. The main function of the charged particle hodoscopes is to provide input to the first level of the NA62 trigger system. The NA48-CHOD, which was used in NA62's predecessor experiment NA48 [8] and is located upstream of LAV12, just after the RICH. The NA48-CHOD consists of two planes of 20 mm thick plastic scintillator slabs, 64 vertical and 64 horizontal which is shown in Figure 16. Each scintillating slab is individually connected to photomultipliers via a light guide. A signal in both planes is required to produce a track time with a resolution of 200 ps. The charged particle rate at the NA48-CHOD was 13 MHz, with the overall rate at 35 MHz mainly resulting from particles produced from interactions with upstream detectors.

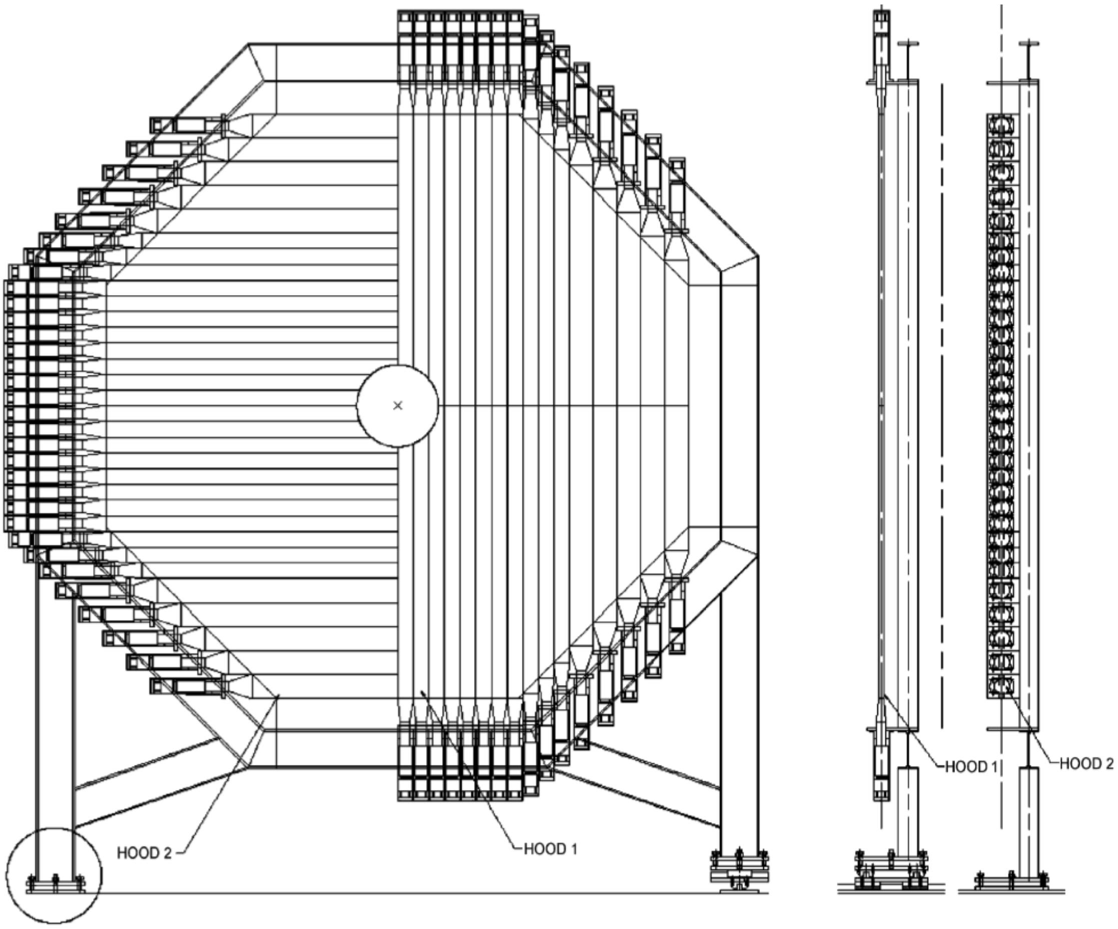

Figure 16: Schematic of the NA48-CHOD [8].

The CHOD is located after the twelfth LAV station, before the LKr and consists of an array of 152 30 mm thick plastic scintillating tiles that cover an acceptance radius of between 140 and 1070 mm, and is shown in Figure 17. The scintillating tiles overlap by 1 mm and are supported by a glass fibre foil (G10). The light produced from each tile is collected with a WLS fibre and is bundled into groups of four, read out by Silicon photomultipliers (SiPM). The overall time resolution of the detector was 1.1 ns. The CHOD sustains a particle rate of 45 MHz, with 13 MHz from charged particles.

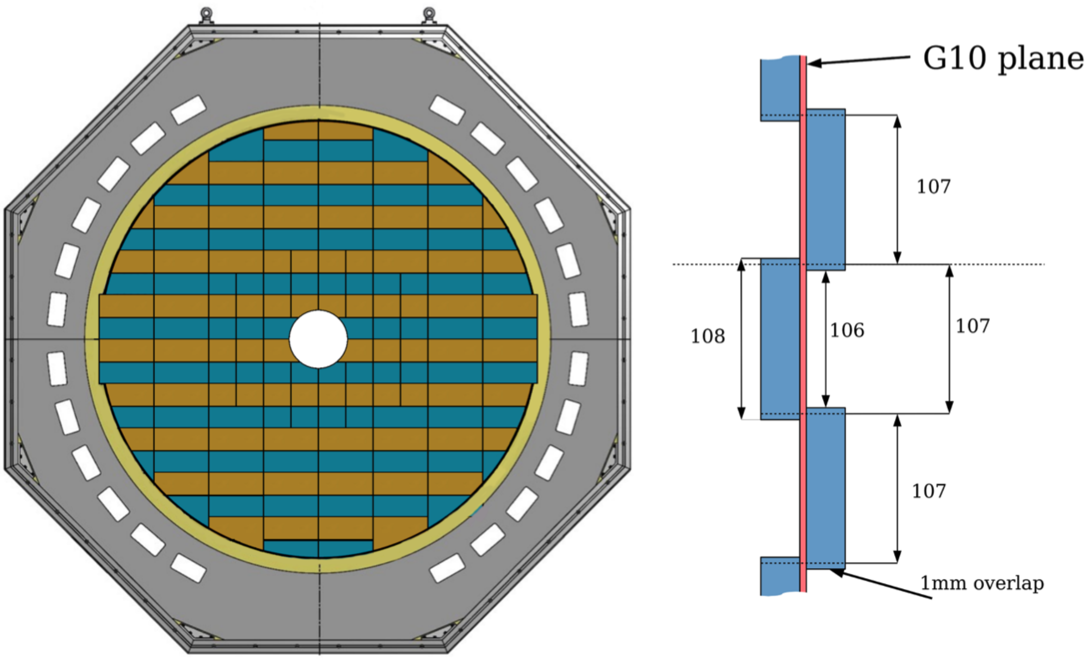

Figure 17: Layout of the NA62 CHOD. Left, the front cross-section of scintillating tiles with the support structure. Right, side cross-section of the scintillating tiles [9].

0.3.8 Muon Veto System

The Muon Veto System (MUV) is used to discriminate between pions and muons in conjunction with the RICH and LKr. The system comprises a hadronic sampling calorimeter made from two modules, MUV1 and MUV2, utilising iron as an absorber material. MUV1 was a newly designed module for NA62, and MUV2 was repurposed from the front of the NA48/2 hadronic calorimeter [8]. MUV1 consists of 24, 26.8 mm thick iron plates with 23 layers of scintillator strips that are 9 mm thick with a width of 60 mm, placed in between, shown in the schematic in Figure 18. The scintillating strips alternate between horizontal and vertical, resulting in 12 horizontal and 13 vertical. The material of MUV1 produces a total thickness of 4.1 X_0. The light that is produced within the strips is collected by WLS fibres and read out by PMTs.

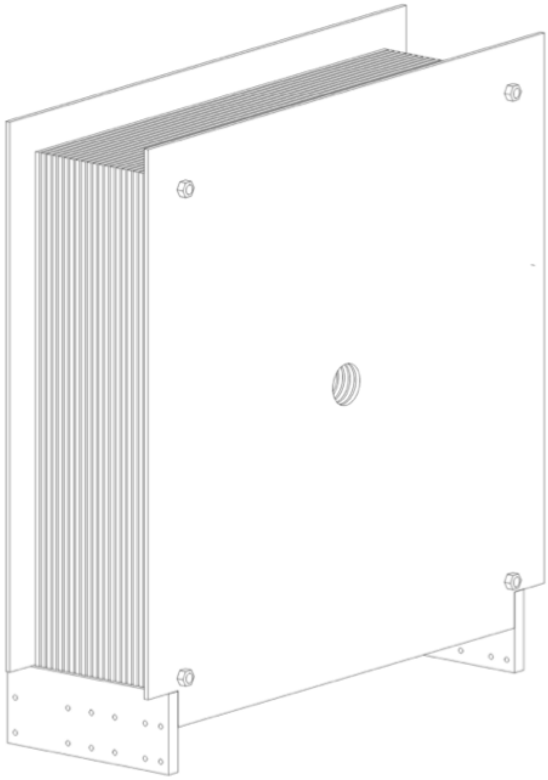

Figure 18: Schematic of the MUV1 module used in the Muon Veto System at NA62.

MUV2 is similar in construction with 2×22 strips per plane versus 48 for MUV1, as shown in Figure 19. MUV2 has thinner iron plates, at 25 mm thick, resulting in a lower total thickness of $3.7\,X_0$.

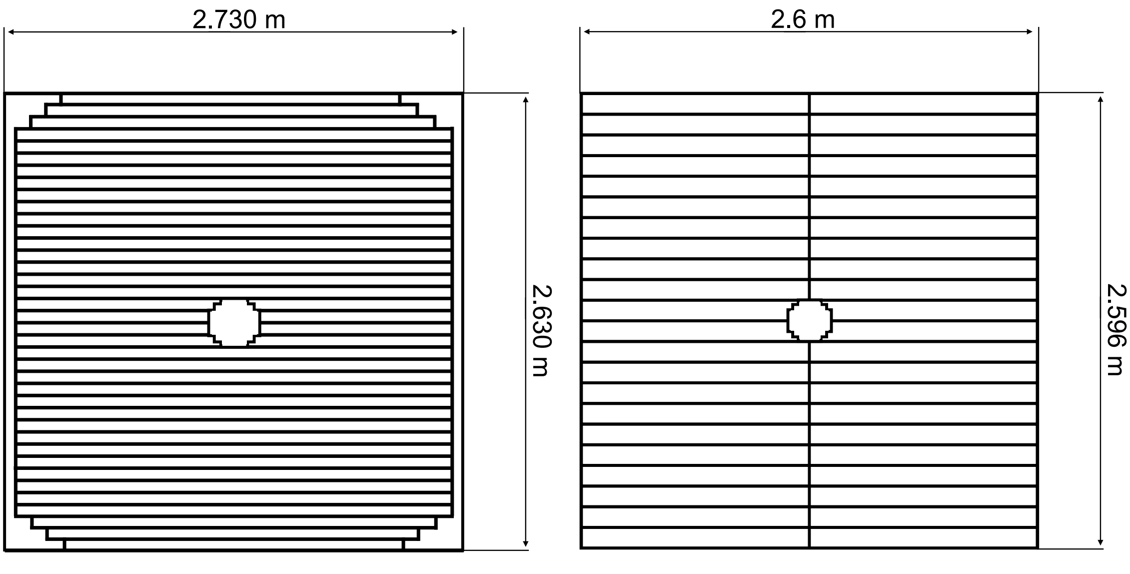

Figure 19: Layout of scintillating tiles for a horizontal layer for: Left, MUV1 and Right, MUV2.

Downstream of the hadronic calorimeter, the MUV3 detector is located behind an 80 cm thick iron wall and is used for muon identification. The MUV3 comprises of 140, 59 mm thick scintillating tiles and eight smaller tiles around the beam pipe to account for the high particle rate from the beam. Each tile is read out directly by two PMTs, without any WLS to improve time resolution of the detector to 0.6 ns. The overall muon rate within the detector for a nominal beam intensity is 13 MHz. The MUV3 is primarily used as a veto within the L0 trigger system to veto events that include a muon.

Additionally, there is a peripheral muon veto detector (MUV0), which is a scintillating hodoscope designed to detect low momentum π^- from the kaon decay $K^+ \to \pi^+\pi^+\pi^-$, which are deflected into the positive X dimension and miss the acceptance of the RICH.

0.3.9 Hadronic Sampling Calorimeter (HASC)

The Hadronic Sampling Calorimeter (HASC) is located downstream of the MUV3 and BEND magnet which deflects positively charged particles in the negative X direction. The HASC is designed to detect π^+ from the kaon decay $K^+ \to \pi^+\pi^+\pi^-$ and comprises of nine modules of 60 lead plates and 60 scintillating layers covering an acceptance of 0.48 m to 0.18 m in X and -0.15 to 0.15 in Y from the beam axis. Each scintillating layer has 10 longitudinal sections which are optically coupled to WLS fibres and read out by SiPMs.

0.3.10 Detectors for Run 2

Between run 1 (2016-2018) and run 2 (2021-present), a number of modifications were applied to the NA62 detector setup shown in Figure 20. An additional GTK station was placed upstream of the first station to reduce tracking inefficiencies. The GTK2 station was also moved upstream of the scraper magnet to minimise the amount of background from beam interactions.

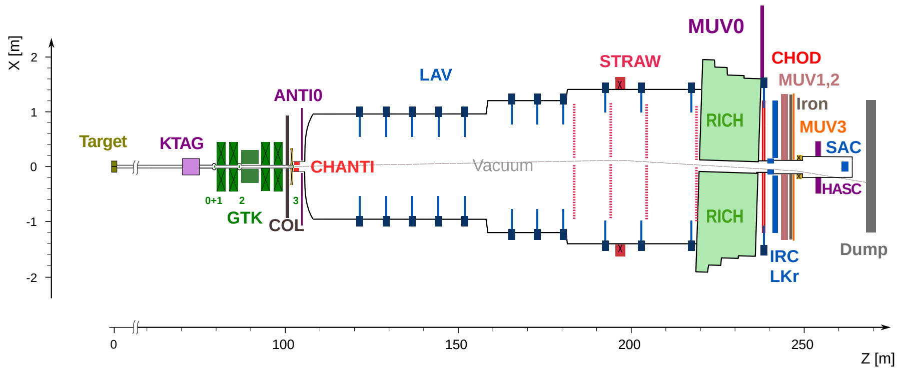

Figure 20: X-Z diagram of NA62 detector setup for Run 2

The VetoCounter (VC) was designed and built to suppress the backgrounds produced by decays of kaons that occur upstream of the fiducial volume. The VC comprises three scintillating stations, with two placed before the final collimator, separated by a lead plate and one after the final collimator. Each station comprises of nine tiles of size 40x120 mm and two 20x120 mm, with the smaller tiles located on either side of the beam pipe and is read out by PMTs.

A veto hodoscope (ANTI-0) was placed just before the entrance to the fiducial volume. The ANTI-0 is designed to veto the halo particles produced from the beam when they enter the FV to reduce the chance of halo particles from being identified as secondary decay products. The detector comprises 280 scintillating tiles, which are then read out by four SiPMs [10].

A second HASC station was also added to the opposite side of the beam to cover the positive X direction and to make the overall station symmetric.

0.4 Trigger and Data Acquisition System (TDAQ)

The high intensity of the NA62 experiment requires a triggering system that can limit the eventual data that is stored while keeping useful data to satisfy the experiment's physics goals. The NA62 trigger system is comprised of two stages: a hardware trigger (L0) and a software trigger (L1), which is also known as the high-level trigger (HLT). Each stage of the trigger system is designed to reduce the incoming rate by a factor of 10. With a rate of O(10 MHz) kaon decays within the fiducial volume, the L0 was designed to reduce this to O(1 MHz) and the L1 then reducing the rate written to storage to O(100 kHz).

To fit with NA62 broad physics goals, the trigger system is designed to be flexible. The L0 trigger involves packets of data sent from fast sub-detectors, which contain the results of the conditions found in Table 2 and is then processed by the L0 Trigger Processor (L0TP). These L0 trigger conditions can be combined to produce a hardware trigger, such as those used in the 2024 run shown in Table 4. The coincidence of the L0 conditions from Table 2 have to be within 6.25 ns of the trigger reference time which is provided by the RICH detector. Once an event passes the L0 trigger other detectors then send information packets to the NA62 PC farm for use in the L1 trigger.

Detector	**Condition**	**Description**
RICH	RICH	At least two signals in the detector
	RICH3	At least three signals in the detector
	RICH8	At least eight signals in the detector
CHOD	Q1	At least one signal in the detector
	H3	At least three signals in the detector
	Q2	At least one signal in each of two different quadrants
	QX	At least one signal in each of two diagonally-opposite quadrants
	UTMC	Fewer than five signals in the detector
NA48-CHOD	NA48-CHOD	At least one signal in the detector
MUV3	M1	At least one signal in the detector
	MO1	At least one signal in the outer tiles
	MO2	At least two signals in the outer tiles
	MOQX	At least one signal in each of two diagonally-opposite quadrants
LKr	E5	At least 5 GeV deposited in the LKr
	E10	At least 10 GeV deposited in the LKr
	E20	At least 20 GeV deposited in the LKr
	E30	At least 40 GeV deposited in the LKr
	E40	At least 40 GeV deposited in the LKr
	C2E5	At least 5 GeV deposited in the LKr by at least two clusters
	LKr40	Logical OR between E40 and C2E5

Table 2: List of L0 trigger conditions for each detector at NA62.

The L1 trigger then uses a number of algorithms for three more detectors, which are described in Table 3. These conditions are then combined to produce a L1 trigger stream shown in Table 4 and then combined with a L0 stream to form a trigger mask. NA62 has a number of trigger masks defined in Table 4 to fulfil its numerous physics goals. For an event to be saved to permanent storage, at least one of these trigger masks has to be accepted. Each mask has a corresponding downscaling, which reduces the rate of events passing that trigger to reduce the amount saved to permanent storage. However, this results in the loss of interesting physics data so the downscaling is minimised as much as possible.

Detector	**Condition**	**Description**
KTAG	KTAG5	Requires hits in at least five sectors within a 5 ns time window
LAV	LAV2-11	Requires at least three signals in LAV stations 2-11 within a 4 ns time window
STRAW	STRAW	Designed for collection of PNN decays, requiring a single positive track originating from the Fiducial Volume (FV), momentum below 50 GeV/c, closest distance of approach (CDA) to the nominal beam axis below 200 mm with a Z position upstream of the first STRAW station
	STRAW-Exo	Requires at least one negatively charged track
	STRAW-ExoTight	Extension of STRAW-Exo requiring the negative track to have momentum below 65 GeV/c, a CDA below 500 mm from the nominal beam axis and a Z position of the CDA of at least 80 m
	STRAW-MMiss	Extension of STRAW-ExoTight, which includes a requirement on the missing mass squared from the track and the nominal beam four-momenta to suppress $K^+ \rightarrow \pi^+\pi^+\pi^-$ decays
	STRAW-MT	Requires at least three tracks originating from the FV
	STRAW-DV5	Requires pairs of tracks originating from a displaced vertex with a CDA at least 50 mm from the nominal beam axis
	STRAW-1TRK	Requires one positively charged track originating in the FV with a momentum less than 65 GeV/c
	STRAW-MUV3	Extension of STRAW to include extrapolation of the track to the acceptance of MUV3 and requiring it to be within $\|x\|\&\|y\| < 1320$ and a radius of 103 mm

Table 3: List of L1 trigger conditions for each detector utilised at NA62.

Trigger	L0 Conditions	L1 Conditions	Downscaling
Normalisation	RICHxRICH3x!M1xQ1	KTAGxSTRAW-1TRK	400
Pnn	RICHxRICH3x!M1x!QXxUTMCx!E40	KTAGx!LAV2-11xSTRAW-MUV3	1
MinBias	RICHxQ1		400
3	RICHxRICH3x!QXxUTMCx!E40	KTAGx!LAV2-11xSTRAW	500
Electron MultiTrack	RICHxRICH8xQXxH3xE20xC2E5	STRAW-MM	1
MultiTrack	RICHxRICH8xQXxH3	KTAGxSTRAW-ExoTight	100
Muon MultiTrack	RICHxRICH8xMO2xQXxH3	KTAGxSTRAW-ExoTight	1
7	RICHxQ1xE30	KTAGxSTRAW	10
8	RICHxRICH8xQ2xE30xC2D5	STRAW-ExoTight	4
Neutrino	!RICH16xMOQXx!Q2xUTMCxE5	KTAGxSTRAW-1TRK	1

Table 4: List of triggers used for the NA62 2024 run, along with the L0 and L1 conditions and the relevant downscaling factors.

A more detailed explanation of the full NA62 trigger system and the resultant performance can be found here [11].

0.5 NA62 Software

The NA62 collaboration maintains an offline software framework (na62fw) which is used for the physics analysis of data collected from the experiment. The software framework comprises of three separate packages:

- NA62MC: A GEANT4 [12] based Monte Carlo simulation of the experiment, used to compare against the data collected.
- NA62Reconstruction: Reconstruction software used to create hits and candidates for all sub-detectors from data collected by the experiment and data from MC simulations.
- NA62Analysis: A software tool kit to conduct physics analysis on the reconstructed data.

Each module of na62fw utilises the tools found within the ROOT analysis framework [13]. A more detailed description of the software framework can be found in Section ??.

0.6 NA62 Results

The latest result from NA62 contains data collected in the periods 2016-2018 and 2021-2022. The data collection during 2016, 2017 and 2018 was at mean intensities of 240, 330 and 400 MHz, respectively. With 2021 and 2022 collected at a mean intensity of 580 MHz. The trigger streams used for the data collection for the $K^+ \to \pi\nu\bar{\nu}$ (PNN) in this period differ from those that are defined in Section 0.4 and are defined below (L0 in red and L1 in blue):

- Normalisation (to collect normalisation sample of $K^+ \to \pi^+\pi^0$):
  RICHxQ1x$\overline{\text{MUV3}}$xKTAGxSTRAW-1TRK

- PNN:
  RICHxQ1x$\overline{\text{MUV3}}$xUTMCx$\overline{\text{QX}}$x$\overline{\text{LKr40}}$xKTAGxSTRAW-1TRKxSTRAWx$\overline{\text{LAV}}$

The selection criteria for the signal and the normalisation involve overlapping conditions, requiring a kaon candidate in the KTAG with a measured momentum from the GTK in the range of 72.7 to 77.2 GeV/c. The $\pi+$ is identified using information from the RICH, LKr, and the Muon Veto System. The π^+ is matched to the kaon candidate by extrapolating the kaon and pion tracks from the GTK and the STRAW, respectively, to form a vertex, defined as the midpoint of the trajectories closest approach.

A signal-specific selection then requires no signals in the LAV downstream of the vertex within 3 ns, no signals in the IRC or SAC within 5 ns and no signals in the LKr within 10 mm of the π^+ impact point. A more detailed breakdown of the selection criteria can be found here [14].

The nominalisation events are selected through the missing mass parameters shown in Figure 1 in the range 0.010 - 0.026GeV^2/c^4 [14], with respect to the π^0 mass [15].

The dataset from run one and run two was combined through a number of methods set out in [14], resulting in the measurement of the branching ratio

$$\mathcal{B}(K^+ \to \pi^+\nu\bar{\nu}) = (13^{+3.3}_{-3.0} \times 10^{-11}) \tag{0.4}$$

The overall dataset from 2016 to 2022 had an expected number of background events of 18^{+3}_{-2} and an observation of 51 signal events [14]. This resulted in a p-value of the background only hypothesis of 2×10^{-7}, corresponding to a rejection of five sigma. Representing the first observation of $K^+ \to \pi\nu\bar{\nu}$.

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